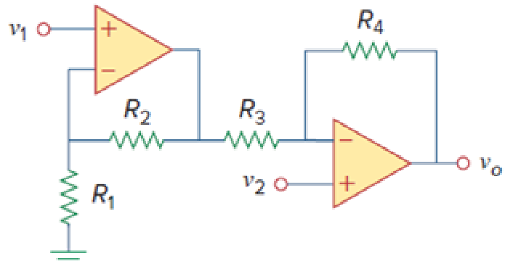
Chapter 5, Problem 87P

Problem

Figure 5.105 displays a two-op-amp instrumentation amplifier. Derive an expression for *vo* in terms of *v*1 and *v*2. How can this amplifier be used as a subtractor?

Figure 5.105:

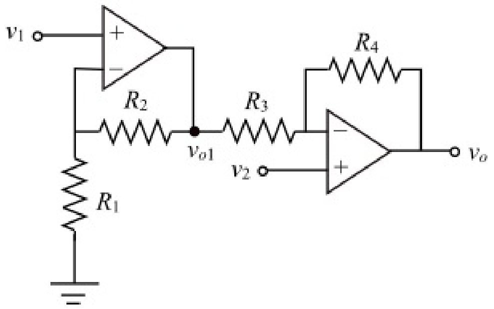


Step-by-step solution

Step 1 of 4

Refer to Figure 5.105 in the text book for the two op-amp instrumentation amplifier circuit.

Redraw the circuit by marking the output voltage of the first op-amp.



According to virtual ground concept, the voltage at the inverting input of the first op-amp is equal to v_1, and the voltage at the inverting terminal of the second op-amp is equal to v_2

Step 2 of 4

Apply Kirchhoff's current law at the inverting terminal of the first op-amp.

$$\frac{v_1}{R_1}+\frac{v_1-v_{o1}}{R_2}=0$$

$$\frac{v_{o1}}{R_2}=\left(\frac{1}{R_1}+\frac{1}{R_2}\right)v_1$$

$$\begin{aligned} v_{o1} &= R_2\left(\frac{1}{R_1}+\frac{1}{R_2}\right)v_1 \\ &= \left(1+\frac{R_2}{R_1}\right)v_1 \end{aligned}$$

Step 3 of 4

Apply Kirchhoff's current law at the inverting terminal of the second op-amp.

$$\frac{v_2 - v_{o1}}{R_3} + \frac{v_2 - v_o}{R_4} = 0$$

$$\left(\frac{1}{R_3} + \frac{1}{R_4}\right)v_2 - \frac{v_{o1}}{R_3} = \frac{v_o}{R_4}$$

Substitute $\left(1+\frac{R_2}{R_1}\right)v_1$ for v_{o1}.

$$\left(\frac{1}{R_3} + \frac{1}{R_4}\right)v_2 - \frac{1}{R_3}\left(1+\frac{R_2}{R_1}\right)v_1 = \frac{v_o}{R_4}$$

$$\begin{aligned} v_o &= R_4\left[\left(\frac{1}{R_3} + \frac{1}{R_4}\right)v_2 - \frac{1}{R_3}\left(1+\frac{R_2}{R_1}\right)v_1\right] \\ &= \left(1+\frac{R_4}{R_3}\right)v_2 - \frac{R_4}{R_3}\left(1+\frac{R_2}{R_1}\right)v_1 \\ &= \left(1+\frac{R_4}{R_3}\right)v_2 - \left(\frac{R_4}{R_3}+\frac{R_2R_4}{R_1R_3}\right)v_1 \end{aligned}$$

Thus, the expression for v_o is $\boxed{\left(1+\frac{R_4}{R_3}\right)v_2 - \left(\frac{R_4}{R_3}+\frac{R_2R_4}{R_1R_3}\right)v_1}$.

Step 4 of 4

To use the amplifier as a subtractor, the gains of the two inputs should be equal.

$$1+\frac{R_4}{R_3} = \frac{R_4}{R_3}+\frac{R_2R_4}{R_1R_3}$$

$$\frac{R_2R_4}{R_1R_3} = 1$$

$$R_2R_4 = R_1R_3$$

$$\frac{R_1}{R_2} = \frac{R_4}{R_3}$$

Write the expression for the output voltage.

$$\begin{aligned} v_o &= \left(1+\frac{R_4}{R_3}\right)v_2 - \left(\frac{R_4}{R_3}+\frac{R_2R_4}{R_1R_3}\right)v_1 \\ &= \left(1+\frac{R_4}{R_3}\right)v_2 - \left(\frac{R_4}{R_3}+1\right)v_1 \\ &= \left(1+\frac{R_4}{R_3}\right)(v_2 - v_1) \end{aligned}$$

(or)

$$v_o = \left(1+\frac{R_1}{R_2}\right)(v_2 - v_1)$$

Thus, the amplifier can be used as a subtractor by making the ratios of the resistors equal, that is, $\boxed{\frac{R_1}{R_2} = \frac{R_4}{R_3}}$ or by making $\boxed{R_1 = R_4}$ and $\boxed{R_2 = R_3}$.

The subtractor has a gain of $\left(1+\frac{R_4}{R_3}\right)$, which is equal to $\left(1+\frac{R_1}{R_2}\right)$.